

Iran R. Roman

Queen Mary University of London
School of Electronic Engineering and Computer Science
London, UK

Phone: +1 (650) 441-4323
Email: i.roman@qmul.ac.uk
Web: iranroman.github.io

Academic Affiliation

2024^{-now} | **Assistant Professor and Lecturer of Artificial Intelligence**, Queen Mary University of London

- Designed and delivered the graduate-level course ECS759P: Artificial Intelligence
- Supervised student research on multimodal artificial intelligence
- Initiated and led an industry research partnership with Meta Inc. on multimodal AI for Project Aria
- Led the Computational Auditory Perception Laboratory

Education

2014-21 | **Ph.D. in Computer-based music theory and acoustics**, Stanford University

Thesis title: Mathematical models of rhythm synchronization and anticipation

Advisors: Dr. Julius O. Smith, Dr. Edward W. Large, and Dr. Chris Chafe

2025-26 | **Postgraduate certificate in academic practice**, Queen Mary University of London

2009-14 | **B.A. in Biology, minor in chemistry**, University of North Texas

2009-14 | **B.M. in Music theory**, University of North Texas

2009-14 | **B.A. in German language**, University of North Texas

Professional Experience

2021-24 | **Human-centered AI researcher**, postdoctoral scholar, New York University, New York, NY

- Optimized large multimodal models (LMMs) to understand 3D scenes and guide human behavior in AR.
- Developed algorithms that use spatial cues to detect, localize and track objects in the real world.
- Contributed to open-source python projects, including librosa, spatialscape, mirdata, and micarraylib.

2020-21 | **Signal processing engineer**, internship, Tesla Inc, Palo Alto, CA

- Designed signal processing algorithms for the embedded audio system of the Model S and Model X.
- Wrote testing pipelines to identify signal distortion in A2DP and PCM streaming.
- Designed signal processing layouts with the AudioWeaver software for dedicated audio hardware.

2018-19 | **Deep learning researcher**, internship, Apple Inc, Cupertino, CA

- Optimized the transformer neural architecture to carry out automatic speech recognition (ASR).
- Collected human data with novel sensors designed for natural voice interaction with Siri.
- Designed deep neural network architectures that integrate speech with multimodal signals.

2017-18 | **Machine learning engineer**, internship, Poly, Santa Cruz, CA

- Developed neural networks and curated datasets for biometric authentication of users with audio headsets.
- Optimized the size of the neural network to be small enough to run in an embedded system.

Publications

Refereed Journal Articles

- Harding EE, Kim JC, Demos A, **Roman IR**, Tichko P, Palmer C, Large EW. Musical neurodynamics. *Nature Reviews Neuroscience*. 2025.
- McGowan E, Rulff J, Castelo S, Wu G, Chen S, Lopez R, Steers B, **Roman IR**, Dias FF, Qian J, Solunke P, Middleton M, McKendrick R, Zhao C, Silva C. Design and implementation of the transparent, interpretable, and multimodal (TIM) AR personal assistant. *IEEE Computer Graphics and Applications*. 2025.
- Castelo S, Rulff J, Solunke P, McGowan E, Wu G, **Roman IR**, Lopez R, Steers B, Sun Q, Bello J, Feest B, Middleton M, McKendrick, Silva C. HuBar: a visual analytics tool to explore human behavior based on fNIRS in AR guidance systems. *Transactions in Visualization and Computer Graphics*. 2024.
- Fuentes M, Salamon J, Zinemanas P, Rocamora M, Paja G, **Roman IR**, Miron M, Serra X, Bello JP. Soundata: Reproducible use of audio datasets. *Journal of Open Source Software*. 2024; 9(98):6634.
- Castelo S, Rulff J, McGowan E, Steers B, Wu G, Chen S, **Roman IR**, Lopez R, Brewer E, Zhao C, Qian J, Cho K, He H, Sun Q, Vo H, Bello J, Krone M, Silva C. ARGUS: visualization of AI-assisted task guidance in AR. *Transactions in Visualization and Computer Graphics*. 2023.
- **Roman IR**, Roman AS, Kim JC, Large EW. Hebbian learning with elasticity explains how the spontaneous motor tempo affects music performance synchronization. *PLOS Computational Biology*. 2023 Jun 7;19(6):e1011154.
- Large EW, **Roman IR**, Kim JC, Cannon J, Pazdera JK, Trainor LJ, Rinzel J, Bose A. Dynamic models for musical rhythm perception and coordination. *Frontiers in Computational Neuroscience*. 2023 May 17;17:1151895.
- Liang BS, Liang AS, **Roman IR**, Weiss T, Duinkharjav B, Bello JP, Sun Q. Reconstructing room scales with a single sound for augmented reality displays. *Journal of Information Display*. 2023 Jan 2;24(1):1-2
- **Roman IR**, Washburn A, Large EW, Chafe C, Fujioka T. Delayed feedback embedded in perception-action coordination cycles results in anticipation behavior during synchronized rhythmic action: A dynamical systems approach. *PLOS Computational Biology*. 2019 Oct 31;15(10):e1007371.
- Washburn A, **Roman IR**, Huberth M, Gang N, Dauer T, Reid W, Nanou C, Wright M, Fujioka T. Musical role asymmetries in piano duet performance influence alpha-band neural oscillation and Behavioral synchronization. *Frontiers in Neuroscience*. 2019 Oct 15;13:1088.
- Huberth M, Dauer T, Nanou C, **Roman IR**, Gang N, Reid W, Wright M, Fujioka T. Performance monitoring of self and other in a turn-taking piano duet: A dual-EEG study. *Social Neuroscience*. 2019 Jul 4;14(4):449-61.
- Jiménez JE, Crego RD, Soto GE, **Roman IR**, Rozzi R, Vergara PM. Potential impact of the alien American mink (*Neovison vison*) on Magellanic woodpeckers (*Campephilus magellanicus*) in Navarino Island, Southern Chile. *Biological Invasions*. 2014 Apr;16(4):961-6.

Refereed Articles in Conference Proceedings

- Carone BJ, **Roman IR**, Ripollés P. Evaluating Multimodal Large Language Models on Core Music Perception Tasks. In *Proceedings of the NeurIPS Workshop on AI for Music*, 2025.
- Shimada K, Politis A, **Roman IR**, Sudarsanam P, Diaz-Guerra D, Pandey R, Uchida K, Koyama Y, Takahashi N, Shibuya T, Takahashi S, Virtanen T, Mitsufuji Y. Stereo Sound Event Localization and Detection with On-screen/offscreen Classification. In *Proceedings of the Workshop on Detection and Classification of Acoustic Scenes and Events (DCASE)*, 2025.
- Roman AS, **Roman IR**, Bello JP. Latent Acoustic Mapping for Direction of Arrival Estimation: A Self-Supervised Approach. In *Proceedings of the IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2025.
- Urrego-Gómez I, Colton S, **Roman IR**. Vibe Sorcery: Integrating Emotion Recognition with Generative Music for Playlist Curation. In *Proceedings of the ISMIR Workshop on Large Language Models for Music & Audio*, 2025.

- Chang A, Li Y, **Roman IR**, Poeppel D. Spectrotemporal modulation: Efficient and interpretable feature representation for classifying speech, music, & environmental sounds. In *Proceedings of Interspeech*, 2025.
- Wang A, **Roman IR**. Toward Affective Empathy in AI: Encoding Internal Representations of “Artificial Pain”. In *Proceedings of the Cognitive Computational Neuroscience (CCN) meeting*, 2025.
- Zimokha M, Jamone L, **Roman IR**. Combining Recurrent & Bayesian Models for Action Anticipation with Multiple Cues. In *Proceedings of the Cognitive Computational Neuroscience (CCN) meeting*, 2025.
- Wang A, Gupta A, **Roman IR**, Zubiaga A. QMUL at SemEval-2025 Task 11: Explicit Emotion Detection with EmoLex, Feature Engineering, and Threshold-Optimized Multi-Label Classification. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval)*, 2025.
- Pedroza HE, Abreu W, Corey R, **Roman IR**. Guitar-TECHS: an electric guitar dataset covering techniques, musical excerpts, chords and scales using a diverse array of hardware. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- Peng X, Chen K, **Roman IR**, Bello JP, Chakravarthula P, Sun Q. Perceptually-guided acoustic "foveation". In *32nd IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)*, 2025.
- Walls K, **Roman IR**, Van Ert K, Harper C, Adu-Gilmore L. Analyzing pitch content in traditional Ghanaian seperewa songs. In *1st Latin American Music Information Retrieval (LAMIR) workshop*, 2024.
- Pedroza HE, Abreu W, Corey R, **Roman IR**. Leveraging electric guitar tones and effects to improve robustness in guitar tablature transcription modeling. In *27th International Conference on Digital Audio Effects (DAFx)*, 2024.
- Roman AS, **Roman IR**, Bello JP. Robust direction of arrival estimation from deep acoustic imaging. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.
- **Roman IR**, Ick C, Sivan D, Roman AS, McFee B, Bello JP. Spatial Scaper: a library to simulate and augment soundscapes for sound event localization and detection in realistic rooms. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.
- Kushwaha SS, **Roman IR**, Fuentes M, Bello JP. Sound source distance estimation in diverse and dynamic acoustic conditions. In *Proceedings of the IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2023.
- **Roman IR**, Faronbi D, Burger-Weiser I, Adu-Gilmore L. F0 analysis of Ghanaian pop singing reveals progressive alignment with equal temperament over the past three decades: a case study. In *Proceedings of the Sound and Music Computing Conference (SMC)*, 2023.
- Faronbi D, **Roman IR**, Bello JP. Exploring approaches to multi-task automatic synthesizer programming. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- Kushwaha SS, **Roman IR**, Bello JP. Analyzing the effect of equal-angle spatial discretization on sound event localization and detection. In *Proceedings of the Workshop on Detection and Classification of Acoustic Scenes and Events (DCASE)*, 2022.
- **Roman IR**, Bello JP. Micarraylib: software for reproducible aggregation, standardization, and signal processing of microphone array datasets. In *Proceedings of the Workshop on Detection and Classification of Acoustic Scenes and Events (DCASE)*, 2021.

Other Non-Refereed Publications

- Bozilovic B., **Roman IR**. Decoding melodic acoustic features from neural data. [2025-AES-AIMLA](#).
- Bhargava S, Chu E, Lee M, Chen C, Walls K, Steers B, **Roman IR**. Generative singing style transfer across genres. [ismir2024program.ismir.net](#)
- Rodríguez Castañeda M, **Roman IR**. The Voice of an instrument: analysis of x-vectors for music applications. [ismir2024program.ismir.net](#)
- Bas E, **Roman IR**. Enhancement of speech and language models through unsupervised Learning with Music Datasets. [ismir2024program.ismir.net](#)

- Walls KL, **Roman IR**, Steers B, Georgieva E. Total variation in popular rap vocals from 2009–2023: extension of the analysis by Georgieva, Ripollés & McFee. ismir2023program.ismir.net
- **Roman IR**, Reid W. An oscillatory neural network model of motor dynamics during continuous periodic movement. cs379c.stanford.edu
- O'Brien T, **Roman IR**. A RNN for musical structure processing and expectation. cs224d.stanford.edu
- **Roman IR**. Assessing neuroplasticity with convolutional and recurrent neural networks. vision.stanford.edu

Open-Source Contributions

Software

- audiblelight, project co-creator and current maintainer, github.com/AudibleLight
- spatialscaper, project co-creator and current maintainer, github.com/SpatialScaper
- musicinformationretrieval, current maintainer of the website and github repo, musicinformationretrieval.com
- seldvisualsynth, project co-creator, github.com/SELDVisualSynth
- librosa, re-wrote the API to efficiently handle multi-channel audio, github.com/librosa
- mirdata, wrote code to load and use the EGFxSet dataset, github.com/mirdata
- micarraylib, created the project and wrote the first API version in its entirety, github.com/micarraylib
- soundata, wrote code to process spatial audio and microphone array datasets, github.com/soundata

Data

- MULTIVOX: spatial audio-visual dataset of singing groups, zenodo.com/multivox
- Guitar-TECHS: an electric guitar dataset covering techniques, musical excerpts, chords and scales using a diverse array of hardware, zenodo.com/guitartechs
- EGSet12: twelve real & original solo electric guitar performances with diverse playing styles to evaluate guitar tablature transcription, zenodo.com/egset12
- EGFxSet: electric guitar tones processed through real effects, zenodo.com/egfxset

Conference Presentations with Refereed Abstracts

Podium

- **Roman IR**, Large EW. Neuronal dynamics of strong anticipation in musical action. In *Abstracts of the SIAM Conference on Applications of Dynamical Systems*, 2023
- **Roman IR**, Rabinovitch E, Golumbic EZ, Large EW. A dynamic model of neural oscillation can synchronize with natural speech. In *Abstracts of the 2022 Advances and Perspectives in Auditory Neuroscience meeting*, 2022
- Roman, AS, **Roman, IR**. Individual musician's spontaneous performance rates affect interpersonal synchrony in joint musical performance: a dynamical systems model. In *Abstracts of the biennial meeting of the Society for Music Perception and Cognition*, 2019.
- **Roman IR**, Washburn A, Large E, Chafe C, Fujioka T. Delayed feedback embedded in perception-action coordination cycles results in anticipation behavior. In *Abstracts of the International Conference on Music Perception and Cognition*, 2018.
- **Roman IR**, Fujioka T. Music syntactic processing is influenced by integration of local and global harmonic structures: an ERP study. In *Abstracts of the Cognitive Neuroscience Society meeting*, 2016.

Poster

- **Roman IR**, Rabinovitch E, Golumbic EZ, Large EW. A dynamic model of neural oscillation can synchronize with natural speech. In *Program No. 241.02. San Diego, CA: Society for Neuroscience*, 2022
- **Roman IR**, Fujioka T. Music syntactic processing is determined by the architecture of a recurrent neural network. In *Abstracts of the Neurosciences and Music Conference*, 2017.
- **Roman IR**, Huberth M, Gang N, Dauer T, Reid W, Nanou C, Wright M, Fujioka T. A dual piano performance EEG study: the effect of the partner's animacy and melodic content on alpha-band oscillations. In *Abstracts of the Cognitive Neuroscience Society meeting*, 2017.
- Huberth M, Dauer T, **Roman IR**, Nanou C, Gang N, Reid W, Wright M, Fujioka T. Involvement or irrelevance: representation of the self vs. other in joint piano performance recorded by dual-EEG. In *Abstracts of the Cognitive Neuroscience Society meeting*, 2017.
- **Roman IR**, Imam J, Stearns T. Characterization of five human p53 mutants using the budding yeast (*Saccharomyces cerevisiae*) as a model. In *Abstracts of the Annual Biomedical Research Conference for Minority Students*, 2013.
- **Roman IR**, Jiménez JE, Vergara P, Rozzi R. Magellanic woodpecker (*Campephilus magellanicus*) behavior when approached by humans in the context of ecotourism. In *Abstracts of the Ecological Society of America meeting*, 2013.

Invited Lectures and Seminars

- 2024 | Bridging data, models, and theory: from computational musicology to multimodal AI and neural signal processing. Data Science Seminar Series. University of Oregon. Feb 26th, Eugene, OR
- 2023 | Multimodal sensing to recognize steps in medical procedures. Defense Advanced Research Projects Agency, Perceptual Task Guidance PI Meeting at Wright State University. May 23rd, Dayton, OH
- 2023 | Mathematical models of music anticipation and synchronization. Maastricht University. Mar 20th, Maastricht, Netherlands
- 2022 | Deep learning for music. Universidad de Nuevo León. Dec 9th, Nuevo León, Mexico
- 2022 | Action and object recognition for TIM: a transparent, interpretable, and multimodal personal assistant. Defense Advanced Research Projects Agency, Perceptual Task Guidance PI Meeting at the Massachusetts Institute of Technology. Nov 8th, Cambridge, MA
- 2021 | A transparent, interpretable, and multimodal personal assistant. Defense Advanced Research Projects Agency, Perceptual Task Guidance Kickoff Meeting. Nov 17th, Washington, DC
- 2021 | Hebbian learning with elasticity explains music performance synchronization and speech envelope processing. New York University Center for Neural Science. Oct 11th, New York City, NY
- 2020 | Deep learning applied to audio. Universidad Politécnica de Madrid. Nov 13th, Madrid, Spain
- 2020 | Mathematical models of music anticipation and synchronization. New York University Center for Neural Science. Feb 22nd, New York City, NY
- 2019 | Delayed feedback embedded in perception-action coordination cycles results in anticipation behavior during synchronized rhythmic action: a dynamical systems approach. Stanford University Center for Mind, Brain, and Computation. May 20th, Stanford, CA

Honors and Awards

- 2023 | Honorable mention for best paper, IEEE VIS: Visualization & Visual Analytics
- 2019 | Excellence in advocacy award, Stanford University Diversity and Advocacy Committee
- 2019 | Outstanding artificial intelligence project for Siri speech, Apple Inc
- 2016 | Honorable mention for best paper, Stanford University Deep Learning for Genomics and Biomedicine
- 2013 | Best poster presentation, Annual Biomedical Research Conference for Minority Students
- 2013 | Honorable mention for oral presentation, Stanford Summer Research Program
- 2013 | Nationally competitive awards public recognition, University of North Texas Honors Day

Grants, Fellowships, and Scholarships

2025	Meta Platforms Inc, Unrestricted Academic Gift	\$25,000 USD
2025	UKRI EPSRC, Outreach and Underpinning the Pipeline for AI Skills	£20,000 GBP
2025	The British Computer Society, Implementing Levelling Up Maths for Computer Science	£3,200 GBP
2024	International Conference on Digital Audio Effects, Travel Scholarship	£500 GBP
2024	National Science Foundation, Spatial Sound Scene Description IIS-1955357, REU Supp.	\$17,600 USD
2023	National Science Foundation, Spatial Sound Scene Description IIS-1955357, REU Supp.	\$15,000 USD
2023	SIAM Conference on Applications of Dynamical Systems, Travel Scholarship	\$650 USD
2023	Cold Spring Harbor, Science Forward: Inclusive Excellence Symposium, Travel Scholar	\$250 USD
2021	National Science Foundation, Spatial Sound Scene Description IIS-1955357, REU Supp.	\$45,000 USD
2021	Defense Advanced Research Projects Agency, Perceptually Enabled Task Guidance (proposal co-authored with PI: Claudio Silva, grant HR001121S0015-PTG-FP-041)	\$5,129,870 USD
2019	Stanford University, Human-Centered Artificial Intelligence Research Fellowship	\$70,000 USD
2015	Stanford University, Mind, Brain and Computation Graduate Research Traineeship	\$4,500 USD
2015	National Institute of Mental Health, Cognitive Neuroscience Summer Institute Scholar	~\$2,500 USD
2014	Stanford University, Graduate Fellowship	\$308,000 USD
2014	Howard Hughes Medical Institute, Exceptional Research Opportunities Capstone	~\$20,000 USD
2013	Howard Hughes Medical Institute, Exceptional Research Opportunities Fellowship	~\$20,000 USD
2012	Howard Hughes Medical Institute, Undergraduate Research Fellowship	~\$20,000 USD

Student Research Advisees

Current

2025 ^{-now}	Zorka Bozilovic, Undergrad, Queen Mary University of London
2025 ^{-now}	Ayat Adam, Undergrad, Queen Mary University of London
2024 ^{-now}	Gerardo Meza, Ph.D. candidate, National Autonomous University of Mexico
2024 ^{-now}	Brandon Carone, Ph.D. candidate, New York University
2023 ^{-now}	Kelvin Walls, Ph.D. candidate, New York University
2023 ^{-now}	Hegel Pedroza, Ph.D. candidate, National Autonomous University of Mexico
2022 ^{-now}	Chuyang Chen, Undergrad, New York University

Past

2025	Carlos Hernandez Oliván, Ph.D., University of Zaragoza (now Postdoc at NTT)
2025	Giselle Pedraza, Undergrad, University of Texas at Austin
2024-25	Isabel Urrego Master student, Queen Mary University of London (now Research Engineer at Meta)
2024-25	Angeline Wang, Master student, Queen Mary University of London
2024-25	Mariia Zimokha, Master student, Queen Mary University of London
2024-25	Mariana Rodriguez Castañeda, Master student, National Autonomous University of Mexico
2022-25	Ameyaltzin Castillo Almazán, Ph.D. candidate, National Autonomous University of Mexico
2018-25	Adrian S. Roman, Master student, University of Southern California (now Ph.D. at NYU)
2024	Karinna Cvijanovich, Undergrad, Amherst College (now Master student at Dartmouth)
2023-24	Wallace Abreu, Master student, Federal University of Rio de Janeiro (now Ph.D. at same institution)
2023-24	Clément Sicard, Master student, ETH Zurich (now Data Engineer at Isomorphic Labs)
2023	Lauren Pryor, Undergrad, California Institute of Technology
2023	Nalicha Antoine, High-School, Achievement First Brooklyn High School (now Undergrad at RPI)
2022-23	Saksham Singh Kushwaha, Master student, New York University (now Ph.D. at UT Dallas)
2022	Aidan Singh, Undergrad, New York University (now Software Engineer at Amazon)
2022	Bridget Meier, High-School, Staten Island Technical High School (now Undergrad at NYU)
2021	Aliaa Mahgoub, High-School, Brooklyn Technical High School, (now Undergrad at Yale)

Courses Taught

Graduate level

2024-25 | ECS 759P: Artificial Intelligence, Queen Mary University of London
 2022-23 | CSCI-GA 3813: Advanced Laboratory in Computer Science, New York University
 2022 | Deep Learning for Music Information Retrieval, National Autonomous University of Mexico
 2017-18 | MUSIC 280: Music Department Teaching Assistant Training, Stanford University
 2017 | EE 367A: Signal Processing Models in Musical Acoustics, Stanford University
 2016-17 | MUSIC 320B: Intro to Audio Signal Processing Part II: Digital Filters, Stanford University
 2016 | EE 367D: Signal Processing Techniques for Digital Audio Effects, Stanford University
 2015-16 | MUSIC 320A: Intro to Audio Signal Processing Part I: Spectrum Analysis, Stanford University

Undergraduate level

2014 | BIOL 4022: Microbiology Laboratory, University of North Texas
 2013 | BIOL 1710: Principles of Biology I, University of North Texas
 2012 | BIOL 3452: Genetics Laboratory, University of North Texas

Week-long tutorials

2017-25 | Deep Learning for Music Information Retrieval, Stanford University (once a year)
 2020 | AI for Audio using Neural Networks, Mexican Center for Music and Sonic Arts
 2020 | Artificial Intelligence, Stanford University Summer Pre-Collegiate Institute
 2018 | Introduction to Deep Learning, Higher Technological Institute of Southern Guanajuato
 2016 | Mathematics for Spectral Analysis and Digital Filters, Mexican Center for Music and Sonic Arts
 2016 | Artificial Intelligence, Stanford University Summer Pre-Collegiate Institute

Online (on demand)

- Generative Music AI Course, (collaboration with Universitat Pompeu Fabra), [The Sound of AI](#)
- Deep Learning for Music Information Retrieval: Generative Models, [dl4genaudio.github.io](#)
- Artificial Intelligence for Audio Using Neural Networks, [cmmas.com](#)

Service

Editorial Roles

2025 | Guest Editor, *Frontiers in Computational Neuroscience*, Research Topic: "Artificial Neuroscience: machine cognition and behavior"

Article Review

2025-26 | Annals of the New York Academy of Sciences
 2025 | Neural Information Processing Systems (NeurIPS)
 2025 | European Signal Processing Conference (EUSIPCO)
 2025 | Public Library of Science (PLOS), Computational Biology
 2025 | Journal of the Audio Engineering Society
 2024 | Transactions of the International Society for Music Information Retrieval
 2024 | IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
 2024 | Signal Processing: Image Communication
 2023-24 | IEEE International Conference on Acoustics, Speech, and Signal Processing
 2023-24 | IEEE International Workshop on Machine Learning for Signal Processing
 2023-24 | eLife
 2023-24 | Music & Science
 2021-24 | International Society for Music Information Retrieval
 2023 | Journal of Attention, Perception, & Psychophysics
 2022 | Cognitive Systems Research Journal

Grant Review

2021-22 | National Research Project Competition, Chile's National Science Foundation

Doctoral Thesis Committees

2025 | Marco Comunità, Queen Mary University of London
 2025 | Daniel Krauze, Tampere University
 2025 | Carlos Hernández Oliván, University of Zaragoza

Academic

2025 | Meeting chair, Machine Learning for Time Series Analysis II, IEEE ICASSP
 2022-25 | Volunteer professor, National Autonomous University of Mexico
 2023 | Meeting chair, Workshop on Neuroscience and Neural Networks, National College of Mexico
 2022 | Meeting chair, IEEE Mexico LatAm BISH Bash
 2021-22 | Policy council officer, Northside Center for Child Development
 2019-20 | ASSU executive director of graduate student affordability, Stanford University
 2017-19 | Member of the working group for diversity and inclusion, Stanford University Department of Music
 2017-19 | Seminar coordinator, Stanford University Center for Mind, Brain, and Computation
 2012-13 | Student ambassador, University of North Texas World Languages and Literatures

Other

2013 | Bilingual tour guide, UNESCO Cape Horn Biosphere Reserve, Puerto Williams, Chile

Diversity and Inclusion

2025-26 | Program Director, Leveling Up: Math for Computer Science, Queen Mary University of London

- Directed an outreach program supporting 20 high school students from London public schools.
- Students prepared to confidently take the A-level Math exams and pursue university STEM degrees.

2023-25 | Mentor, IEEE SPS Mentoring Experiences for Underrepresented Young Researchers

- Mentored talented undergraduate & graduate students to gain research experience in signal processing & AI.
- Students published their work as peer-reviewed conference papers at conferences like DAFx and ICASSP.
- Students also gained grant writing experience, each one fundraising \$4,000 USD to travel to conferences.

2022-25 | Mentor, Women in Music Information Retrieval (WiMIR)

- Introduced aspiring researchers to the field of Music Information Retrieval (MIR).
- Met individually with each student to discuss how to navigate career options in industry and academia.
- Mentored students to gain hands-on skills carrying out MIR research.

2016-24 | Scholarship coordinator, Diversity in Music Technology Summer Scholarship, Stanford University

- Prepared and publicized a summer tuition scholarship application to study music technology at Stanford.
- This scholarship has supported students from backgrounds and nationalities underrepresented in STEM.

2021-24 | Recruiter, Undergrad Research Experience, Music and Audio Research Lab, New York University

- Publicized REU positions in our lab at minority-serving engineering programs across the United States.
- Produced a pool of talented applicants with strong representation from minority groups in STEM.

2021-23 | Research mentor, Applied Research Innovations in Science and Engineering, New York University

- Mentored high-school students carrying out summer research at the Music and Audio Research Lab.
- Helped students prepare a research presentation, delivered at a symposium celebrating minorities in STEM.

2023 | Presentation, Science Forward: Towards Inclusive Excellence Symposium, Cold Spring Harbor Lab

- Presented the poster titled "Highlighting representation in multi-modal machine learning research".
- Created a spotlight around BIPOC researchers by inviting them to present at high-impact venues.
- Recorded their presentations to curate a [YouTube playlist](#) (with their consent).

2016 | Program leader, Stanford Summer Research Program, Stanford Medical School

- Mentored a group of diverse undergraduate students carrying out summer research.
- Guided students to prepare high-quality oral and poster presentations for a research symposium.
- Provided students with feedback on their personal statements for Ph.D. program applications.

2015 | Program leader, Stanford Biosciences ADVANCE Summer Institute, Stanford Medical School

- Planned and conducted a journal club series with a diverse group of new Ph.D. students.

Languages

- Spanish: Native
- English: Fluent
- German: Advanced

Press

- 2025 | [Music Resonates And Rewards The Brain](#), *Forbes*, May 19th
- 2025 | [When Your Favorite Song Plays, Your Brain ‘Physically Embodies’ Music](#), *Science Shows*, *Study Finds*, May 9th
- 2025 | [Groundbreaking research reveals how the brain turns sound into music](#), *Queen Mary EECS Research News*, Apr 8th
- 2024 | [New tool helps analyze pilot performance and mental workload in AR](#), *NYU Research News*, Oct 14th
- 2019 | [Delayed neural communication may underlie anticipatory behaviors](#), *EurekAlert!*, Oct 31st
- 2019 | [Stanford’s Human-Centered AI Institute awards 30 seed grants](#), *The Stanford Daily*, May 1st
- 2016 | [High school students wrestle with perennial questions and make connections between art practice and other disciplines](#), *Stanford University News*, Aug 9th
- 2013 | [UNT undergraduate student earns Howard Hughes grant](#), *The North Texan*, Jun 5th

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